

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_diversity_failure_replay.
- Best TSPLIB holdout gap: phase6_diversity_failure_replay.
- Best synthetic holdout gap: phase6_no_replay.

Run Metadata

- run_name: run_20260519_010943_i
- started_at_local: 2026-05-19 01:09:43
- finished_at_local: 2026-05-19 01:29:04
- duration_hhmm: 00:19
- duration_seconds: 1160.547
- seed_offset: 8000
- replicate_label: i
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.152008	0.0	0.096732	0.236772	0.158253	0.0	0.629021	0.76	0.034627
phase6_random_replay	random	score_only	False	0.101948	0.0	0.064876	0.236772	0.157262	0.19457	0.638472	0.76	0.060473
phase6_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.171206	0.348017	0.261637	0.913465	0.56	-0.027264

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.157489	0.003941	0.101653	0.236772	0.172033	0.19457	0.798354	0.76	0.044768
phase6_stratified_random_replay	stratified_random	score_only	False	0.174134	0.0	0.110813	0.236772	0.163224	0.221162	0.0	0.76	0.0
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.213387	0.064916	0.159398	0.236772	0.214847	0.130039	0.894426	0.56	-0.027845
phase6_residual_failure_replay	residual_failure	score_only	False	0.098826	0.014284	0.068083	0.038798	0.122645	0.194445	0.869774	0.76	0.099436
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.084911	0.0	0.054034	0.171206	0.266037	0.175969	0.658181	0.76	0.097602
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.158373	0.010343	0.104544	0.171206	0.269051	0.172748	0.0	0.76	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.152008, synthetic 0.0, combined 0.096732.
- Accepted-epoch count 2, mean accepted code novelty 0.629021, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.158253, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency 0.034627 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.152008 across 7 instances; family means: ch=0.135771, kroD=0.074152, pcb=0.177813, pr=0.223347, rd=0.201643, st=0.115556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3174, "expected_gap": 0.40822, "family": "a", "name": "a280", "optimality_gap": 0.23071, "residual_gap": -0.17751}, {"best_known_cost": 14379, "cost": 17655, "expected_gap": 0.324988, "family": "lin", "name": "lin105", "optimality_gap": 0.227832, "residual_gap": -0.097156}],

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{"best_known_cost": 629, "cost": 757, "expected_gap": 0.257552, "family": "eil", "name": "eil101", "optimality_gap": 0.203498, "residual_gap": -0.054054}}
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phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.101948, synthetic 0.0, combined 0.064876.
- Accepted-epoch count 2, mean accepted code novelty 0.638472, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.157262, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.133654, mean selected residual gap -0.011002.
- Adaptation efficiency 0.060473 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.101948 across 7 instances; family means: ch=0.078705, kroD=0.157979, pcb=0.226653, pr=0.056214, rd=0.039823, st=0.075556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3055, "expected_gap": 0.403955, "family": "a", "name": "a280", "optimality_gap": 0.184568, "residual_gap": -0.219387}, {"best_known_cost": 629, "cost": 702, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.116057, "residual_gap": -0.140223}, {"best_known_cost": 14379, "cost": 15411, "expected_gap": 0.325433, "family": "lin", "name": "lin105", "optimality_gap": 0.071771, "residual_gap": -0.253662}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.913465, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.261637.
- Replay selection diversity 0.177669, mean selected expected gap 0.308783, mean selected residual gap 0.066974.
- Adaptation efficiency -0.027264 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.403722, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.002404}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.136407}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.323861, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.004952}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.157489, synthetic 0.003941, combined 0.101653.

- Accepted-epoch count 2, mean accepted code novelty 0.798354, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.172033, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.13247, mean selected residual gap 0.011597.
- Adaptation efficiency 0.044768 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.157489 across 7 instances; family means: ch=0.144398, kroD=0.158261, pcb=0.187207, pr=0.213889, rd=0.160936, st=0.093333.
- Panel synthetic_holdout mean gap 0.003941 across 4 instances; family means: clustered_gaussian=0.015764, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3487, "expected_gap": 0.402947, "family": "a", "name": "a280", "optimality_gap": 0.352074, "residual_gap": -0.050873}, {"best_known_cost": 629, "cost": 850, "expected_gap": 0.253418, "family": "eil", "name": "eil101", "optimality_gap": 0.351351, "residual_gap": 0.097933}, {"best_known_cost": 14379, "cost": 18111, "expected_gap": 0.321511, "family": "lin", "name": "lin105", "optimality_gap": 0.259545, "residual_gap": -0.061966}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.174134, synthetic 0.0, combined 0.110813.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.163224, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.237473, mean selected residual gap -0.012027.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.174134 across 7 instances; family means: ch=0.17463, kroD=0.226026, pcb=0.180945, pr=0.273172, rd=0.164349, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3356, "expected_gap": 0.402171, "family": "a", "name": "a280", "optimality_gap": 0.30128, "residual_gap": -0.100891}, {"best_known_cost": 629, "cost": 806, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.281399, "residual_gap": 0.025119}, {"best_known_cost": 14379, "cost": 18147, "expected_gap": 0.326838, "family": "lin", "name": "lin105", "optimality_gap": 0.262049, "residual_gap": -0.064789}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.894426, and final complexity 0.56.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.214847, size bias 8.3e-05, and failure concentration 0.130039.
- Replay selection diversity 0.266358, mean selected expected gap 0.168427, mean selected residual gap 0.025262.

- Adaptation efficiency -0.027845 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.402947, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.001629}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.137043}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.326977, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": 0.001836}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.098826, synthetic 0.014284, combined 0.068083.
- Accepted-epoch count 2, mean accepted code novelty 0.869774, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.038798, mean hardness 0.122645, size bias 0.029507, and failure concentration 0.194445.
- Replay selection diversity 0.077595, mean selected expected gap 0.258116, mean selected residual gap 0.061398.
- Adaptation efficiency 0.099436 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.098826 across 7 instances; family means: ch=0.052307, kroD=0.068846, pcb=0.120466, pr=0.18483, rd=0.107838, st=0.105185.
- Panel synthetic_holdout mean gap 0.014284 across 4 instances; family means: clustered_gaussian=0.015764, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 7542, "cost": 10207, "expected_gap": 0.240918, "family": "berlin", "name": "berlin52", "optimality_gap": 0.353355, "residual_gap": 0.112437}, {"best_known_cost": 2579, "cost": 3401, "expected_gap": 0.405273, "family": "a", "name": "a280", "optimality_gap": 0.318728, "residual_gap": -0.086545}, {"best_known_cost": 629, "cost": 752, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.195548, "residual_gap": -0.066455}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.084911, synthetic 0.0, combined 0.054034.
- Accepted-epoch count 3, mean accepted code novelty 0.658181, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.266037, size bias 0.120322, and failure concentration 0.175969.
- Replay selection diversity 0.198205, mean selected expected gap 0.305588, mean selected residual gap -0.034026.
- Adaptation efficiency 0.097602 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.084911 across 7 instances; family means: ch=0.063492, kroD=0.109843, pcb=0.192583, pr=0.034523, rd=0.087484, st=0.042963.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3201, "expected_gap": 0.405351, "family": "a", "name": "a280", "optimality_gap": 0.241179, "residual_gap": -0.164172}, {"best_known_cost": 629, "cost": 700, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.112878, "residual_gap": -0.154213}, {"best_known_cost": 14379, "cost": 15974, "expected_gap": 0.321761, "family": "lin", "name": "lin105", "optimality_gap": 0.110926, "residual_gap": -0.210835}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.158373, synthetic 0.010343, combined 0.104544.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.269051, size bias 0.120322, and failure concentration 0.172748.
- Replay selection diversity 0.196929, mean selected expected gap 0.300716, mean selected residual gap -0.024669.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.158373 across 7 instances; family means: ch=0.117164, kroD=0.181366, pcb=0.229863, pr=0.247053, rd=0.11378, st=0.102222.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3343, "expected_gap": 0.405739, "family": "a", "name": "a280", "optimality_gap": 0.296239, "residual_gap": -0.1095}, {"best_known_cost": 14379, "cost": 18560, "expected_gap": 0.321274, "family": "lin", "name": "lin105", "optimality_gap": 0.290771, "residual_gap": -0.030503}, {"best_known_cost": 629, "cost": 743, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.18124, "residual_gap": -0.082671}]

Judge Appendix

Summary of Replay-Aware TSP Benchmark Suite Results

Overview

- **Best holdout TSPLIB gap**: **phase6_diversity_failure_replay** (mean gap 0.0849)
- **Best synthetic holdout gap**: **phase6_no_replay** (mean gap 0.0)
- **Best transfer gap (combined)**: **phase6_diversity_failure_replay** (mean gap 0.0540)
- Total conditions: 9, including variants with and without replay, and variants of replay modes.

Transfer Performance (Primary Evidence)

Condition	Mean T SPLIB Gap	Mean Sy nthetic Gap	Mean Transfer Gap	Replay Mode	Selecti on Mode	Archive Diversit y	Archive Hardne ss	Replay Failure Concentration
phase6_diversity_failure_r eplay	0.084 911	0.0	0.0540 34	diversity_ failure	score_ only	0.17471 8	0.31958 3	0.175969

Condition	Mean T SPLIB Gap	Mean Sy nthetic Gap	Mean Transfer Gap	Replay Mode	Selecti on Mode	Archive Diversit y	Archive Hardne ss	Replay Failure Concentration
phase6_no_replay	0.1520 08	0.0	0.09673 2	none	score_ only	0.24017 9	0.17931 6	0.0
phase6_random_replay	0.1019 48	0.0	0.06487 6	random	score_ only	0.24017 9	0.18779 4	0.19457
phase6_random_replay_co mpression	0.1574 89	0.003941	0.10165 3	random	novelty _gate	0.24017 9	0.19465 7	0.19457
phase6_failure_replay	0.2133 87	0.064916	0.15939 8	failure	score_ only	0.17471 8	0.34902	0.261637
phase6_stratified_random_ replay	0.1741 34	0.0	0.11081 3	stratified_ random	score_ only	0.24017 9	0.18976 8	0.25
phase6_diversity_weighted_ replay	0.2133 87	0.064916	0.15939 8	diversity_ weighted	score_ only	0.24017 9	0.21236 1	0.174603
phase6_residual_failure_re play	0.0988 26	0.014284	0.06808 3	residual_f ailure	score_ only	0.15519 1	0.34213 1	0.194445
phase6_diversity_failure_r eplay_compression	0.1583 73	0.010343	0.10454 4	diversity_ failure	novelty _gate	0.17471 8	0.34021 8	0.170068

Interpretations:

- ****phase6_diversity_failure_replay**** leads in TSPLIB holdout and transfer gaps, showing best generalization to real TSPLIB and synthetic instances.
- ****phase6_no_replay**** yields zero synthetic holdout gaps but worse gaps on TSPLIB, indicating lack of transfer robustness.
- Replay modes involving failure cases (diversity_failure, failure_replay) generally improve transfer gap compared to no replay or random replay.
- Compression-aware replay (random_replay_compression, diversity_failure_replay_compression) slightly reduces final archive diversity but does not improve transfer gap compared to diversity_failure_replay.

Replay Archive Characteristics & Mechanism Analysis

Condition	Arch ive Size	Archive Descriptor Diversity	Archive Hardne ss	Archive Size Bias	Replay Failure Concentration	Replay Selection Diversity	Adaptatio n Efficiency
phase6_diversity_failure_r eplay	30	0.1747	0.3196	0.125	0.17597	0.198	0.098
phase6_no_replay	28	0.2402	0.1793	0.0	0.0	0.0	0.035
phase6_random_replay	30	0.2402	0.1878	0.0	0.19457	0.262	0.060
phase6_failure_replay	31	0.1747	**0.349 0**	**0.125* *	**0.262**	0.178	-0.027
phase6_stratified_random_ _replay	30	0.2402	0.1898	0.0	0.2517	0.187	0.0
phase6_diversity_weighte d_replay	31	0.2402	0.2124	0.0	0.1300	0.266	-0.028

Condition	Archive Size	Archive Descriptor Diversity	Archive Hardness	Archive Size Bias	Replay Failure Concentration	Replay Selection Diversity	Adaptation Efficiency
phase6_residual_failure_replay	28	0.1552	0.3421	0.061	0.1944	0.078	0.099
phase6_random_replay_compression	30	0.2402	0.1947	0.0	0.19457	0.262	0.045
phase6_diversity_failure_replay_compression	31	0.1747	0.3402	0.125	0.1701	0.197	0.0

- Replay archive diversity is highest without selection pressure or diversity weighting (e.g., no_replay and random replay).
- Failure replay methods concentrate replay failures (high failure concentration ~0.17-0.26), indicating replay is focused on tough cases.
- Adaptation efficiency is negative for failure replay and diversity_weighted replay, suggesting less efficient learning despite failure focus.
- Residual_failure_replay achieves the highest adaptation efficiency (0.099) and maintains moderate transfer and TSPLIB gaps.
- Compression pressure lowers archive diversity but does not clearly improve transfer performance.

Code Novelty & Complexity

Condition	Last Code Novelty	Mean Code Novelty	Mean Complexity
phase6_diversity_failure_replay	0.90	0.66	0.76
phase6_no_replay	0.63	0.63	0.76
phase6_random_replay	0.64	0.64	0.76
phase6_failure_replay	0.90	0.91	0.56
phase6_stratified_random_replay	0.0	0.0	0.76
phase6_diversity_weighted_replay	0.87	0.66	0.56
phase6_residual_failure_replay	0.87	0.87	0.76
phase6_random_replay_compression	0.80	0.80	0.76
phase6_diversity_failure_replay_compression	0.0	0.0	0.76

- Conditions with improved transfer gaps (e.g., diversity_failure replay) do not systematically show code novelty gains when compression or novelty gating is used.
- Stratified random replay and diversity_failure replay compression show zero code novelty but moderate transfer gaps.
- Algorithmic invention is not strongly indicated by lexical code novelty, matching the rule to prioritize deterministic metrics.
- Complexity varies modestly; failure replay conditions have lower complexity (0.56) but higher archive hardness.

Concentration and Diversity of Failure Replay

- Failure replay conditions show higher replay failure concentration (0.17-0.26), indicating focused replay around challenging failures.
- Diversity-weighted replay achieves decent replay selection diversity (~0.20-0.27) along with moderate failure concentration.
- Residual failure replay shows lower failure concentration (~0.10-0.19) and lower selection diversity, which may affect transfer.

Key Insights

- **Best Transfer:** phase6_diversity_failure_replay achieves the best transfer gap (0.054) and lowest TSPLIB gap (0.085), indicating effective transfer.
- **No Replay Baseline:** phase6_no_replay has zero synthetic gap but higher TSPLIB gap (0.152), showing limited transfer ability without replay.
- **Failure Replay vs Random Replay:** Failure replay increases archive hardness and failure concentration but does not always improve adaptation efficiency and transfer compared to random replay.
- **Residual Failure Replay:** Yields good adaptation efficiency and low transfer gap (~0.068), with slightly lower archive diversity.
- **Code Novelty:** Declines with compression and novelty gate selection, but transfer gaps can improve or remain stable, indicating code novelty is not a proxy for transfer quality.
- **Diversity and Hardness:** The best transfer condition balances archive diversity (~0.17) and hardness (~0.32), favoring inclusion of difficult instances with moderate diversity.

Conclusion and Recommendations

- Prefer **phase6_diversity_failure_replay** as the benchmark condition due to its superior transfer and held-out TSPLIB optimality gaps.
- The **no replay** condition, while achieving perfect synthetic gaps, underperforms on real TSPLIB transfer, marking replay as beneficial.
- Failure-focused replay methods improve transfer but may reduce adaptation efficiency; balancing replay diversity and failure concentration is critical.
- Compression or novelty gating can reduce code novelty without harming transfer, suggesting that reducing lexical novelty is not detrimental.
- Recommend deeper mechanism analysis on replay failure concentration and diversity to optimize replay buffer construction.
- Do not conflate code novelty with algorithmic innovation: effective transfer was achieved where code novelty varied, supporting focus on deterministic transfer and gap metrics.